

Robust Short News Classification under Surface Noise and Spurious-Cue Perturbations

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Abstract

This project studies robust short news classification on the AG News dataset. I compare three levels of methods: a majority-class baseline, a TF-IDF + Logistic Regression baseline, and a fine-tuned DistilBERT classifier. I evaluate these models on clean data, a noisy test set with surface-level perturbations, and a targeted adversarial subset with injected World-style cue sentences. On the full clean test set, DistilBERT achieves the strongest performance, reaching 94.57% accuracy, compared with 91.99% for TF-IDF + Logistic Regression. Under surface-level noise, both models show only modest degradation. Under the targeted adversarial setting, both models degrade further, but TF-IDF + Logistic Regression is affected more strongly than DistilBERT. Error analysis shows that DistilBERT produces fewer cases that flip from correct to incorrect under adversarial perturbation. Overall, the results suggest that a lightweight pre-trained transformer is not only stronger than traditional feature-based baselines on standard short news classification, but also more robust to the specific spurious cue attack considered here.

1 Introduction

News topic classification is a standard task in natural language processing. Given an input text sequence $x = (w_1, w_2, \dots, w_n)$, the goal is to predict a label y from a fixed set of categories. In this project, I study a four-class version of the task using the labels *World*, *Sports*, *Business*, and *Sci/Tech*.

This problem is important because text classification is widely used in information retrieval, content organization, and recommendation systems. It is also a useful benchmark for comparing traditional feature-based methods with pretrained transformer models. At the same time, short news texts can be difficult to classify robustly because

they contain limited context and may be sensitive to small changes in formatting, punctuation, or spelling.

A clean-set evaluation alone does not fully capture this issue. A short headline may still be easy for a human to understand after capitalization changes, punctuation removal, or minor typos, but a classifier may become less reliable under such perturbations. In addition, a model might rely on spurious semantic cues rather than the main topic of the article. For example, adding an irrelevant sentence about foreign officials or diplomats may make a non-World article appear more World-like to the model, even though its true label should remain unchanged.

The main question of this project is whether simple preprocessing and lightweight transformer fine-tuning can support robust short-text classification. To answer this question, I compare simple baselines with DistilBERT on both clean and perturbed versions of AG News (Zhang et al., 2015). I first test robustness to surface-level noise, and then extend the evaluation with a targeted adversarial setting based on spurious World-style cue injection.

2 Related Work

The AG News dataset is a standard benchmark for topic classification and was introduced in prior work on character-level neural text classification (Zhang et al., 2015). In modern NLP, pretrained transformer models have become a dominant approach for text classification, with BERT (Devlin et al., 2019) serving as a central reference point. DistilBERT (Sanh et al., 2019) provides a smaller and faster alternative while retaining much of BERT’s performance, making it a practical choice for an individual course project.

This project is also related to prior work on lightweight text augmentation and perturbation.

For example, EDA proposes simple text augmentation operations for classification tasks (Wei and Zou, 2019). My work is not a direct reproduction of EDA, but it is similarly motivated by the idea that low-cost text modifications can affect classifier behavior. All transformer experiments were implemented with the Hugging Face Transformers framework (Wolf et al., 2020).

3 Methods

I study three levels of methods.

3.1 Majority-class baseline

The simplest baseline is a majority-class classifier that always predicts the most frequent label in the training set. This provides a minimal reference point for the task.

3.2 TF-IDF + Logistic Regression

The stronger traditional baseline is a TF-IDF + Logistic Regression pipeline. The model uses unigram and bigram TF-IDF features extracted from the input text, followed by logistic regression for classification. This method is efficient, easy to implement, and provides a strong non-neural baseline for comparison.

3.3 DistilBERT

The main neural model is DistilBERT (Sanh et al., 2019), specifically `distilbert-base-uncased`. I fine-tune it for four-way classification on AG News. DistilBERT was chosen because it is smaller and faster than larger transformer models such as BERT (Devlin et al., 2019), while still providing strong classification performance.

3.4 Surface-noise perturbation

To evaluate robustness under realistic surface-level corruption, I construct a noisy version of the AG News test set. The perturbations include random capitalization changes, partial punctuation removal, extra spaces, and a limited number of synthetic typos. These perturbations are intended to preserve the original topic label while making the input slightly noisier.

3.5 Targeted adversarial perturbation

Based on instructor feedback, I also construct a targeted adversarial subset. This subset contains only *Sports*, *Business*, and *Sci/Tech* examples. For each example, I inject one short World-style cue

sentence, such as a sentence mentioning government officials, foreign observers, or diplomats. The goal is to test whether the model relies on spurious semantic cues that may increase the probability of predicting *World* even when the original topic remains unchanged.

4 Experimental Setup

4.1 Dataset

I use AG News as the main benchmark (Zhang et al., 2015). The dataset is well suited for this project because it is widely used, manageable in size, and contains clearly defined topic labels. The full test split is used for clean and noisy evaluation. For DistilBERT training, I split the original training set into a 90% training portion and a 10% validation portion, yielding 108,000 training examples and 12,000 validation examples. The standard AG News test set contains 7,600 examples.

4.2 Evaluation metrics

I evaluate models using Accuracy, Macro-F1, and Per-class F1. Accuracy measures overall performance, while Macro-F1 gives equal weight to each category. Per-class F1 helps identify whether some categories are more sensitive to perturbation than others. For the targeted adversarial subset, I also track the *World prediction rate*, since the attack is designed to inject World-style cues.

4.3 Training setup

For TF-IDF + Logistic Regression, I use unigram and bigram TF-IDF features with a fixed feature cap and standard optimization. For DistilBERT, I fine-tune the model for two epochs using AdamW with a learning rate of 2×10^{-5} , batch size 16, and maximum sequence length 128. All transformer experiments were implemented with Hugging Face Transformers (Wolf et al., 2020). DistilBERT training and evaluation were run on Boston University’s SCC cluster using a single GPU, and the model completed training in a short course-project timescale.

5 Results

Table 1 reports overall performance on the full clean and noisy AG News test sets. Table 2 reports performance on the targeted adversarial subset and its corresponding clean subset. Table 3 summarizes adversarial error analysis by counting exam-

ples that were classified correctly on the clean subset but incorrectly after cue injection.

On the full test set, DistilBERT achieved the strongest performance in both the clean and noisy settings. On the clean test set, it reached 94.57% accuracy, outperforming TF-IDF + Logistic Regression by about 2.6 percentage points. Under surface-level noise, both models showed modest degradation. TF-IDF + Logistic Regression dropped by 0.63 points in accuracy, while DistilBERT dropped by 0.71 points. These results suggest that DistilBERT provides stronger overall performance, but that both models remain somewhat sensitive to formatting and spelling perturbations.

The targeted adversarial setting reveals a different pattern. On the corresponding clean subset, TF-IDF + Logistic Regression achieved 92.39% accuracy and DistilBERT achieved 94.39%. After World-style cue injection, TF-IDF + Logistic Regression dropped to 90.23%, while DistilBERT dropped to 93.49%. Thus, the adversarial perturbation reduced performance for both models, but the effect was larger for TF-IDF + Logistic Regression (2.16-point accuracy drop) than for DistilBERT (0.90-point drop).

For DistilBERT, the World prediction rate increased from 1.23% on the clean subset to 2.19% on the adversarial subset. This indicates that the injected World-style cues did create some targeted confusion, but did not cause the model to collapse into predicting World for a large fraction of examples. In addition, because the adversarial subset contains no true World examples, the four-class Macro-F1 values in Table 2 are somewhat artificially depressed and should not be interpreted in the same way as the full-test Macro-F1 values.

The error analysis in Table 3 shows the same overall trend. TF-IDF + Logistic Regression produced 145 examples that were classified correctly on the clean subset but incorrectly on the adversarial subset, whereas DistilBERT produced only 81 such examples. Qualitatively, some adversarial errors did shift examples into the World class, which is consistent with the attack design, but others shifted into *Business* or *Sports* instead. This suggests that the injected cues increased broader cross-class confusion rather than causing only a simple collapse into World predictions.

The majority baseline produced 0 performance on the adversarial subset because that subset excludes true World examples, while the majority

classifier predicts a single class from the original training distribution. As a result, it is not especially informative in this targeted adversarial setting, but it is retained for completeness.

6 Discussion

The results suggest three main findings. First, DistilBERT is consistently stronger than the simpler baselines on standard news classification. On the full clean test set, it achieved the best overall accuracy and Macro-F1, outperforming TF-IDF + Logistic Regression by a clear margin. This indicates that even a lightweight pretrained transformer can provide substantial gains over traditional feature-based methods for short news classification.

Second, surface-level noise affects both TF-IDF + Logistic Regression and DistilBERT, but the impact is relatively modest in the current setting. The noisy test set caused small drops for both models, which suggests that realistic perturbations such as capitalization changes, punctuation removal, extra spaces, and a limited number of typos do reduce reliability, but do not fully break either classifier.

Third, the targeted adversarial setting provides a more informative robustness test than the noisy setting alone. When World-style cue sentences were injected into Sports, Business, and Sci/Tech examples, both models degraded relative to the corresponding clean subset. However, TF-IDF + Logistic Regression was affected more strongly than DistilBERT. This suggests that DistilBERT is not only better on standard classification, but also somewhat more robust to spurious semantic cues.

At the same time, the adversarial results also show that the current attack is limited. For DistilBERT, the World prediction rate increased only modestly, which means that the injected cues created some targeted confusion without causing a strong collapse into World predictions. In addition, the error analysis shows that not all adversarial mistakes shifted into World. Some examples instead shifted into Business or Sports, suggesting that cue injection increased broader cross-class confusion rather than producing only one simple failure mode.

There are also several limitations in the current study. The adversarial examples were generated with a simple template-based injection method rather than a more flexible LLM-based rewriting strategy. Because of this, some injected sentences may be unnatural or only weakly connected to the

Model	Clean Acc.	Noisy Acc.	Acc. Drop	Clean Macro-F1	Noisy Macro-F1	Macro-F1 Drop
Majority Baseline	0.2500	0.2500	0.0000	0.1000	0.1000	0.0000
TF-IDF + Logistic Regression	0.9199	0.9136	0.0063	0.9197	0.9134	0.0063
DistilBERT	0.9457	0.9386	0.0071	0.9457	0.9386	0.0071

Table 1: Performance on the full clean and noisy AG News test sets. DistilBERT is strongest overall, while both TF-IDF + Logistic Regression and DistilBERT show small drops under surface-level noise.

Model	Clean-Subset Acc.	Adv. Acc.	Acc. Drop	Clean-Subset Macro-F1	Adv. Macro-F1	Macro-F1 Drop	Clean World Pred. Rate	Adv. World Pred. Rate
Majority Baseline	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	—	—
TF-IDF + Logistic Regression	0.9239	0.9023	0.0216	0.7000	0.6914	0.0086	—	—
DistilBERT	0.9439	0.9349	0.0090	0.7122	0.7088	0.0034	0.0123	0.0219

Table 2: Performance on the targeted adversarial subset with injected World-style cues, compared against the corresponding clean subset. The adversarial subset contains only Sports, Business, and Sci/Tech examples, so four-class macro-F1 should be interpreted with caution in this setting.

Model	Flip Count	Robustness
TF-IDF + Logistic Regression	145	Lower
DistilBERT	81	Higher

Table 3: Adversarial error analysis. “Flip Count” denotes the number of examples that were classified correctly on the clean subset but incorrectly on the adversarial subset.

original text. In addition, the adversarial subset excludes true World examples, which makes four-class Macro-F1 less informative in that setting. For this reason, the adversarial results are best interpreted through accuracy drops, World prediction rate, and clean-correct to adversarial-wrong error counts rather than Macro-F1 alone.

Overall, the experiments suggest that preprocessing and lightweight transformer fine-tuning are useful for robust short news classification, but that robustness should be evaluated along multiple dimensions. Surface-level noise and spurious semantic cues reveal different weaknesses, and a model that performs well on the clean test set may still exhibit targeted failure patterns under adversarial perturbation.

7 Conclusion

This project studied robust short news classification on AG News using a majority baseline, TF-IDF + Logistic Regression, and DistilBERT. The results showed that DistilBERT was the strongest model overall on both the clean and noisy test sets. Surface-level noise caused only small performance drops, but the targeted adversarial setting revealed a clearer robustness gap between traditional feature-based methods and the transformer model.

On the adversarial subset, both TF-IDF + Logistic Regression and DistilBERT degraded relative to the corresponding clean subset, but TF-IDF + Logistic Regression was affected more strongly. Error analysis further showed that DistilBERT produced fewer cases that flipped from correct to incorrect under adversarial cue injection. These findings suggest that DistilBERT is not only more accurate overall, but also more robust to the specific spurious cue attack considered here.

For future work, a stronger extension would be to generate more natural adversarial examples with an LLM and to test whether adversarial augmentation during training can improve robustness further. More detailed per-class analysis and additional perturbation types could also help clarify which kinds of shortcuts the models rely on most heavily.

8 Code Availability

The source code for this project is available at:

<https://github.com/ThomasFYD/robust-short-news-classification/tree/code>

9 Author Contributions

Yide Fang designed the project, implemented the baselines and DistilBERT experiments, constructed the noisy and adversarial evaluation sets, ran the experiments, analyzed the results, and wrote the report.

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